

John Bolt

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Work Experience

Software Development Engineer II

Apr. 2025 – Present

Amazon, Internal Data Streaming

Detroit, MI

- Designed and implemented encryption and data partitioning features for an internal, managed streaming service based on Apache Flink, supporting >1,000,000 TPS with on-the-fly data transformations and easy configuration by non-developers.
- Replaced a legacy API service and library written for interacting with it with a single, 600-line Java class and hundreds of regression tests, eliminating the service's most common source of bugs, decreasing our infrastructure and operational footprints, and removing tens of thousands of lines of code.
- Fixed integration tests that had not been passing for close to a year, codified their testing infrastructure with IaC, reduced their runtime from several hours to just 5 minutes, and expanded testing coverage to all core use cases.
- Comprehensively rewrote, updated, and consolidated every customer-facing wiki for the product, spanning dozens of out-of-date pages written over almost a decade, into a single user guide that explicitly targeted Data Engineers and Data Scientists as our customers.

Software Development Engineer

Sep. 2022 – Mar. 2025

AWS Business & Cost Management, Model Hosting & Execution

New York City, NY

- Led Cost Anomaly Detection's [root cause improvement project](#), the largest change to the product since its launch, working across the full stack and with data science to deliver the feature on a compressed timeline and with stringent cost constraints for AWS re:Invent.
- Refactored a backend service to decrease costs from horizontal scale and dramatically increase vertical scaling, enabling Cost Anomaly Detection to run on customer payloads 20x larger than before and the product to scale from 27,000 to 3 million customer cost monitors.
- Delivered a project that had been delayed for months with previous ownership and was only partially implemented, designing on the fly, fixing dozens of substantial bugs across 4 interdependent services, producing key metrics for new ML models from a massive end-to-end test, and lowering projected costs by \$1,000,000 annually.
- Designed and implemented a testing framework that duplicates production traffic to evaluate the performance of improved ML models in near-prod environments across multiple interdependent micro-services, contributing to the design and release of four new models.
- Designed two summer intern projects, managed tasks and timelines for junior engineers, and provided mentorship through regular meetings and focused working sessions.

Personal Projects

RFx Optimization Agent

- Designed and wrote an agentic LLM service for addressing communication bottlenecks in RFx processes, collating requirements from these communications, and using them to solve optimization and other problems that result in measurable business impact.
- Created a reusable plugin interface for hooking the agent up to different communication channels such as email, Slack, Discord, etc.
- Leveraged vector databases, knowledge stores, cyclic agentic loops with LangGraph, recursive language models with REPL support for long context management, and mixture of experts to improve LLM performance at the orchestration layer.

Python library for generative architecture in Blender

- Developed a Python library with a fluent interface for constructing 3D architectural primitives in Blender (**AaC: Architecture as Code**).
- Designed all primitives to intelligently handle joins (e.g. join these 10 walls with a floor and this set of windows).
- Created an object-relative interface for specifying the location of new objects, enabling the recursive creation of architectural features as singular, reusable objects.

Education

University of Michigan

May 2022

Bachelor of Computer Science & Political Science

Ann Arbor, MI

Honors/Awards: three terms University Honors, Critical Language Scholarship for Mandarin Study.

Course Highlights: Advanced Algorithms, Operating Systems, Machine Learning, Data Mining, Research Methods.

Volunteering

Writer & Data Analyst

May 2025 – Present

Strong Towns

Detroit, MI

- Wrote and edited the organization's blog posts, such as its recent [zoning explainer](#).
- Performed data analysis to support lobbying efforts targeting the city's mayor, city council, and the Governor's office.
- Managed all tech adoption within the org and worked to instill project management best-practices into the culture.

Skills and Interests

Languages: Java, TypeScript, Python, C/C++, Julia, Rust, R

Technologies: AWS (CloudFormation, CDK), Apache Flink, System Design, Machine Learning, Technical Writing

Spoken Languages: English (native), Mandarin (conversational, diminished over the years)